EXPERIMENT 3:

AIM:

To sort the jobs dataset in descending order based on the job title.

INPUT:

job_id	**job_title**	**min_salary**
IT_PROG	Programmer	4000
HR_REP	HR Representative	3500
MK_MAN	Marketing Manager	9000
FI_ACC	Accountant	5000

PROGRAM:

```
exp3.py > ...
import pandas as pd
df = pd.read_excel(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp3_data.xlsx")
sorted_df = df.sort_values(by="job_title", ascending=False)
print(sorted_df)
```

OUTPUT:

```
> & C:\Users\dhars\AppData\Loc
\Python313\python.exe "c:/Users/dhars/Documents/Query processing/lab_questions/exp3.py"
    job_id          job_title  min_salary
0  IT_PROG         Programmer        4000
2   MK_MAN  Marketing Manager        9000
1   HR_REP  HR Representative        3500
3   FI_ACC         Accountant        5000
```

RESULT:

The details of the jobs were successfully displayed in **descending order of the job title** using the sort_values() function in Pandas.

EXPERIMENT 4:

AIM:

To create a line graph showing the historical closing prices of Alphabet Inc. between two dates.

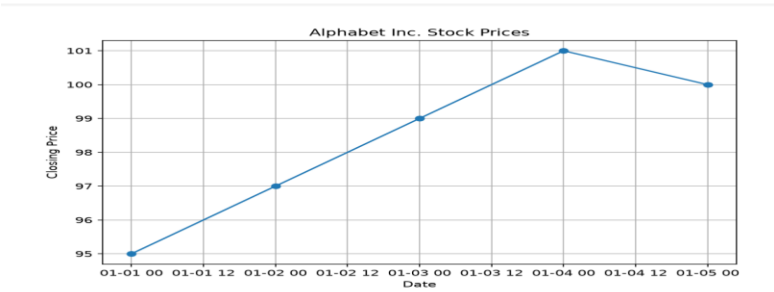
INPUT:

Date	Close	Volume
2023-01-01	95	1000000
2023-01-02	97	1200000
2023-01-03	99	1100000
2023-01-04	101	1300000
2023-01-05	100	1250000

PROGRAM:

```python
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_excel(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp 4,5,6_data.xlsx")
df["Date"] = pd.to_datetime(df["Date"])
start_date = "2023-01-01"
end_date = "2023-01-05"
data = df[(df["Date"] >= start_date) & (df["Date"] <= end_date)]
plt.figure(figsize=(8,5))
plt.plot(data["Date"], data["Close"], marker='o')
plt.title("Alphabet Inc. Stock Prices")
plt.xlabel("Date")
plt.ylabel("Closing Price")
plt.grid(True)
plt.show()
```

OUTPUT:



RESULT:

The historical stock prices of **Alphabet Inc.** between the specified dates were successfully plotted as a **line graph**, showing the variation in closing stock prices over time.